

Comparative Evaluation of Conditioning Modalities using ControlNet

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Abstract

Modern diffusion models support multiple conditioning modalities, including text prompts, edge maps, and semantic segmentation masks. However, it remains unclear how much useful scene information each modality preserves and how effectively diffusion models utilize that guidance. In this project, we treat image reconstruction not as an end goal, but as a controlled benchmark to isolate and quantify the information capacity of different modalities. Given a target image from COCO, we extract sparse text, dense text, Canny edges, segmentation maps, and multi-modal combinations to evaluate how accurately Stable Diffusion 1.5 and ControlNet can restore the original scene using perceptual, semantic, and structural similarity metrics. Furthermore, we extend this evaluation to downstream image modification tasks, providing a definitive reference guide on which specific modalities best balance style adherence with structural preservation when executing targeted edits.

1 Background and Setup

1.1 Problem Statement

Diffusion models such as Stable Diffusion can generate images from text prompts, while ControlNet extends this capability by allowing additional spatial conditioning signals such as Canny edges and semantic segmentation maps. These modalities encode different types of scene information. Text captures object semantics and relationships, edge maps preserve local structure and boundaries, and segmentation maps preserve object-level layout.

Our project asks:

How much useful scene information does each conditioning modality preserve, and how effectively can Stable Diffusion utilize it?

Rather than treating image reconstruction as an end goal, we utilize it as a controlled benchmark to isolate and quantify the information capacity of different modalities. By extracting or writing conditioning signals from a target image and comparing the generated output against the original scene, we measure the exact semantic and structural guidance each representation provides.

Additionally, we use these insights to look beyond pure reconstruction, evaluating how these modalities perform during downstream modification to provide a definitive reference guide on which specific signals users should leverage for targeted editing tasks (e.g., restyling a scene into a "cozy style").

1.2 Inputs and Outputs

The input to our system is a target image from the COCO 2017 validation dataset [Lin et al.(2014)]. From each target image, we construct several possible conditioning inputs:

- sparse text caption,
- dense text caption,
- Canny edge map,
- semantic segmentation map,
- dense text + Canny edge map,
- dense text + semantic segmentation map,
- dense text + Canny edge map + semantic segmentation map.

The output is a reconstructed image generated by Stable Diffusion 1.5 or Stable Diffusion 1.5 with ControlNet.

1.3 Goals and Constraints

Our goal is not to perfectly reproduce the original image pixel-by-pixel. Instead, we want to determine which conditioning modalities best preserve scene information that is useful for reconstruction. A successful method should produce images that are semantically aligned with the target, structurally similar to the target, and perceptually close to the target.

In addition to comparing conditioning modalities, we also investigate how generation parameters influence reconstruction quality. Specifically, we evaluate the effects of classifier-free guidance (CFG) scale and ControlNet conditioning scale. We aim to determine an effective combination of modalities and generation settings that can serve as a practical reference configuration for diffusion-based image reconstruction. By identifying these optimal configurations, this research also unlocks potential applications in extreme, ultra-low bitrate image compression. Because textual captions, Canny edges, and segmentation maps require drastically less storage space than dense pixel data, proving that they preserve sufficient information allows them to act as highly compact scene representations that can be accurately restored or edited downstream without transferring heavy files.

Important constraints include:

- all models are pretrained and frozen,
- experiments are limited to 512 by 512 image generation,
- Stable Diffusion 1.5 uses a CLIP text encoder with a limited context window (77 tokens),
- all results must be evaluated using a consistent benchmark and metric suite.

1.4 Hypothesis

We hypothesize that spatial conditioning modalities will preserve structure better than text-only conditioning, while dense text will preserve semantic information better than sparse text. We further hypothesize that combining dense text with a spatial modality will improve reconstruction quality because the two sources of information are complementary.

1.5 Crux of the Problem

The main challenge is that each modality loses different information. Sparse text removes spatial relationships, dense text is limited by the CLIP token window, Canny edges preserve boundaries but discard object labels and color, and segmentation masks preserve semantic layout but discard texture. The project therefore requires a controlled experimental setup that can compare these modalities fairly.

2 Approach

2.1 Dataset Construction

We use the COCO 2017 validation dataset. To construct a more informative evaluation set, we ranked candidate images using a heuristic based on object count, category diversity, spatial distribution, and Canny edge density. From this ranking, we selected 200 images for the final experiments.

For each selected image, we store the original image, COCO metadata, sparse text prompt, dense text prompt (around 50 words due to limited text tokens), Canny edge map, and semantic segmentation map.

2.2 Text Conditioning

Sparse text prompts are short object-level descriptions intended to preserve only the main semantic entities, directly from the COCO dataset captions. Dense text prompts are manually expanded descriptions that include objects, spatial relationships, scene context, and visual attributes. Because Stable Diffusion 1.5 uses a CLIP text encoder with a limited token context, the dense captions are shortened to remain within the effective prompt length.

2.3 Spatial Conditioning

For Canny conditioning, we compute an edge map from the target image using Canny edge detection [Canny(1986)]. This modality preserves object contours and local geometric structure but removes texture, color, and semantic labels.

For segmentation conditioning, we generate a semantic segmentation map from the target image. This modality preserves object-level region layout and semantic grouping, but removes fine texture and detailed appearance information.

2.4 Models

We use Stable Diffusion 1.5 [Rombach et al.(2022)] for text-only reconstruction. For spatial conditioning, we use Stable Diffusion 1.5 with ControlNet [Zhang et al.(2023)]. The Canny ControlNet is used for edge-map conditioning, and the segmentation ControlNet is used for semantic layout conditioning. All model weights are frozen; our work focuses on evaluation rather than training.

2.5 Experimental Conditions

We evaluate the following seven configurations:

Table 1: Conditioning configurations evaluated in this project.

Configuration	Text	Canny	Segmentation
Baseline Semantic	Sparse	–	–
Advanced Semantic	Dense	–	–
Baseline Structural A	Empty	Yes	–
Baseline Structural B	Empty	–	Yes
Multi-modal A	Dense	Yes	–
Multi-modal B	Dense	–	Yes
Multi-modal C	Dense	Yes	Yes

2.6 Hyperparameter Studies

In addition to comparing conditioning modalities, we evaluate the effect of classifier-free guidance (CFG) scale. CFG controls how strongly Stable Diffusion follows the text prompt during sampling: lower values allow the model prior to dominate, while higher values enforce stronger prompt adherence but may reduce realism or introduce artifacts. We test CFG values of 3.0, 5.0, 7.5, and 10.0 to cover weak, moderate, default-like, and strong text guidance.

For ControlNet experiments, the ControlNet conditioning scale controls how strongly the generated image follows the spatial condition, such as a Canny edge map or segmentation mask. This parameter is independent from CFG: CFG affects text guidance, while ControlNet scale affects spatial guidance. We first report the main modality comparison at ControlNet scale 1.0, then perform a ControlNet-scale sweep to study whether stronger spatial conditioning improves reconstruction.

2.7 Implementation

We implemented batch generation scripts for all text-only, Canny-ControlNet, and segmentation-ControlNet settings. Each script loads the same selected image list, generates the required conditioning signal, runs the corresponding diffusion pipeline, saves the generated image, and evaluates it against the target image.

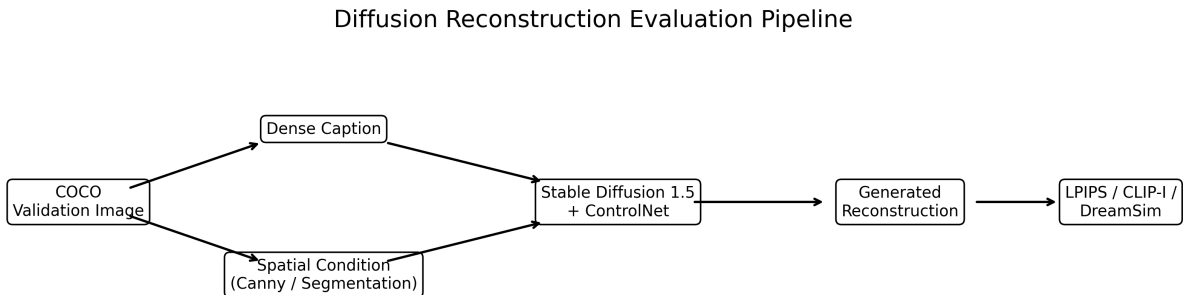


Figure 1: Overview of the reconstruction evaluation pipeline. A COCO image is converted into textual and spatial conditioning signals, used to guide Stable Diffusion 1.5 and ControlNet, and evaluated using LPIPS, CLIP-Score, and DreamSim.

3 Evaluation and Results

3.1 Definition of Success

We define success as the ability of a conditioning modality to guide generation toward an image that is close to the target. Since no single metric fully captures image similarity, we evaluate each output using multiple complementary metrics.

3.2 Metrics

We use the following metrics:

- **LPIPS** [Zhang et al.(2018)]: measures perceptual distance using deep visual features. Lower is better.
- **CLIP-Score** [Hessel et al.(2021)]: measures semantic similarity using deep CLIP embeddings. Higher is better.
- **DreamSim** [Fu et al.(2023)]: measures perceptual and mid-level visual similarity. Lower is better.

3.3 Experimental Setup

All experiments are run on the selected 200-image COCO subset. Each image is resized to 512 by 512 for generation. We use fixed model checkpoints and the same evaluation pipeline across all configurations. For CFG sweeps, we evaluate multiple guidance strengths while keeping the dataset and model settings fixed.

All experiments were executed locally on an Apple Silicon Mac (M4 Pro chip with 48 GB unified memory) using the PyTorch MPS (Metal Performance Shaders) backend to accelerate inference due to limited compute resources.

3.4 Quantitative Results

Table 2: Main reconstruction results averaged over the 200-image evaluation set (CFG = 7.5; ControlNet scale = 1.0).

Method	LPIPS ↓	CLIP-Score ↑	DreamSim ↓
Sparse Text	0.763	24.10	0.687
Dense Text	0.733	33.82	0.501
Canny Only	0.545	28.15	0.432
Segmentation Only	0.671	23.86	0.583
Dense + Canny	0.517	34.44	0.328
Dense + Segmentation	0.641	34.28	0.414
Dense + Canny + Segmentation	0.501	34.33	0.321

Overall Reconstruction Performance. Table 2 compares all six conditioning configurations at CFG = 7.5. Among single-modality approaches, dense text substantially improves semantic reconstruction over sparse text, increasing CLIP-Score from 24.10 to 33.82 while reducing DreamSim

from 0.687 to 0.501. This indicates that richer textual descriptions provide significantly stronger semantic guidance than object-category lists alone.

Among structural modalities, Canny edge conditioning clearly outperforms semantic segmentation conditioning. Canny-only reconstruction achieves LPIPS of 0.545 and DreamSim of 0.432, compared with 0.671 and 0.583 for segmentation-only reconstruction. This suggests that precise object boundaries and geometric structure provide stronger reconstruction cues than coarse semantic regions.

Combining modalities further improves reconstruction fidelity. Incorporating dense text with either spatial modality yields substantial gains, but the Dense + Canny configuration consistently dominates Dense + Segmentation across perceptual metrics. Among double-modality approaches, Dense + Canny achieves the strongest baseline performance (LPIPS: 0.517, DreamSim: 0.328, CLIP-Score: 34.44).

However, our evaluation of the triple-modal configuration (Dense + Canny + Segmentation) reveals that segmentation still provides valuable complementary information when edge maps and text are already present. Provided the ControlNet scale remains moderate, integrating segmentation as a secondary spatial branch alongside Canny improves perceptual metrics even further, pushing LPIPS down to 0.501 and DreamSim to 0.321.

Together, these results support our hypothesis that semantic and structural conditioning are highly complementary, demonstrating that the synergistic combination of dense text, geometric boundaries, and region-level layouts produces the strongest overall reconstruction signal.

3.5 Guidance Scale Sweep

We also evaluate the effect of classifier-free guidance scale. This helps determine whether stronger text guidance improves reconstruction or causes the model to over-prioritize the prompt at the expense of visual structure.

Table 3: CFG sweep evaluation (mean over 200 images; ControlNet scale = 1.0).

CFG	Sparse Text			Dense Text			Dense + Canny			Dense + Segmentation		
	LPIPS ↓	CLIP-Score ↑	DreamSim ↓	LPIPS ↓	CLIP-Score ↑	DreamSim ↓	LPIPS ↓	CLIP-Score ↑	DreamSim ↓	LPIPS ↓	CLIP-Score ↑	DreamSim ↓
3.0	0.745	23.83	0.675	0.728	32.76	0.503	0.518	33.35	0.340	0.641	33.62	0.421
5.0	0.754	23.88	0.682	0.727	33.57	0.497	0.513	34.06	0.326	0.640	34.35	0.415
7.5	0.763	24.10	0.687	0.733	33.82	0.501	0.517	34.44	0.328	0.641	34.28	0.414
10.0	0.768	24.16	0.686	0.738	33.84	0.500	0.528	34.44	0.335	0.647	34.45	0.419

Effect of CFG Guidance Strength. Across all configurations, classifier-free guidance produces relatively modest changes compared to the choice of conditioning modality itself. For sparse text conditioning, increasing CFG slightly improves semantic similarity but generally worsens LPIPS and DreamSim, suggesting that stronger guidance encourages the model to follow the prompt while sacrificing visual fidelity.

Dense text exhibits similar behavior. The best LPIPS and DreamSim values occur at moderate CFG values between 3 and 5, while CLIP-Score continues to increase slightly at higher guidance strengths. This indicates a trade-off between semantic adherence and perceptual similarity.

For multimodal configurations, Dense+Canny remains the strongest performer across all tested CFG values. The differences between CFG settings are small relative to the performance gap

between modalities. This observation suggests that selecting an effective conditioning modality is considerably more important than aggressively tuning CFG.

Overall, the results indicate that moderate CFG values between 5 and 7.5 provide a reasonable balance between semantic guidance and reconstruction fidelity for Stable Diffusion 1.5.

3.6 ControlNet Scale Sweep

We additionally evaluated the effect of the ControlNet conditioning scale while fixing CFG at 7.5. ControlNet scale determines how strongly the generated image follows the spatial conditioning signal. Larger values enforce stronger adherence to edges or segmentation masks, while smaller values allow the diffusion prior to play a larger role.

Table 4: Best ControlNet scale for each configuration and evaluation metric. Values in parentheses indicate the corresponding metric score achieved at that scale.

Configuration	Best Scale (LPIPS ↓)	Best Scale (CLIP-Score ↑)	Best Scale (DreamSim ↓)
Empty + Canny	1.5 (0.521)	1.5 (28.2)	1.0 (0.435)
Empty + Segmentation	2.0 (0.656)	1.5 (24.6)	1.5 (0.568)
Dense + Canny	1.5 (0.500)	0.5 (34.4)	1.0 (0.329)
Dense + Segmentation	2.0 (0.624)	1.5 (34.5)	1.0 (0.412)

Across all spatially conditioned configurations, increasing the ControlNet scale generally improved LPIPS but could reduce CLIP-Score, indicating a trade-off between structural fidelity and semantic alignment. For Dense+Canny, LPIPS improved from 0.519 at scale 1.0 to 0.500 at scale 1.5, while CLIP-Score decreased from 34.3 to 33.6. DreamSim remained relatively stable near scales 1.0–1.5.

Overall, scales between 1.0 and 1.5 consistently provided strong performance across metrics, whereas scale 2.0 often introduced diminishing returns and larger semantic degradation.

Table 5: Triple modality branch-scale ablation at CFG 7.5 (200-image means).

Setting	LPIPS ↓	CLIP ↑	DreamSim ↓
Equal (1.0 / 1.0)	0.501	34.33	0.321
Canny-heavy (1.5 / 0.5)	0.489	33.56	0.334
Seg-heavy (0.5 / 1.5)	0.558	34.43	0.359

Across the triple modality configurations, the scale distribution reveals a direct trade-off between structural layout and semantic preservation. Allocating more weight to the Canny branch (Canny-heavy) optimizes structural fidelity, improving LPIPS to 0.489, but reduces semantic alignment (CLIP falls to 33.56). Conversely, favoring the segmentation branch (Seg-heavy) maximizes semantic alignment with a peak CLIP score of 34.43, but severely degrades structural accuracy, pushing LPIPS to 0.558.

Balancing both spatial branches equally (Equal, 1.0 / 1.0) mitigates these deficits, yielding the best overall perceptual similarity with a DreamSim score of 0.321. Based on this trade-off, we select the equal branch-scale configuration of 1.0 / 1.0 for our final recommendation.

3.7 Qualitative Results

To ensure the reliability of our findings, we conducted extensive manual visual inspections across the dataset to verify whether the numerical improvements captured by our perceptual, semantic, and structural metrics truly reflect qualitative accuracy in the generated images.



Figure 2: Qualitative validation of quantitative metrics.

Having established that our quantitative metrics reliably track visual quality, we next perform a direct qualitative analysis to evaluate how different input modalities impact scene reconstruction. Figure 3 presents representative reconstruction examples from an outdoor sports scene (top row) and an indoor office scene (bottom row). These examples illustrate the relative strengths and weaknesses of semantic, structural, and multimodal conditioning.

Table 6: Example captions used for the qualitative evaluation.

ID	Sparse Caption	Dense Caption
12120	A couple of women playing a game of tennis on a tennis court.	Two women compete in a tennis match on a professional court surrounded by spectators and stadium seating.
27620	A laptop sitting on a desk next to a chair and another computer.	A small home office or study area contains a glass-top desk, a laptop computer, a chair, and additional electronic equipment.

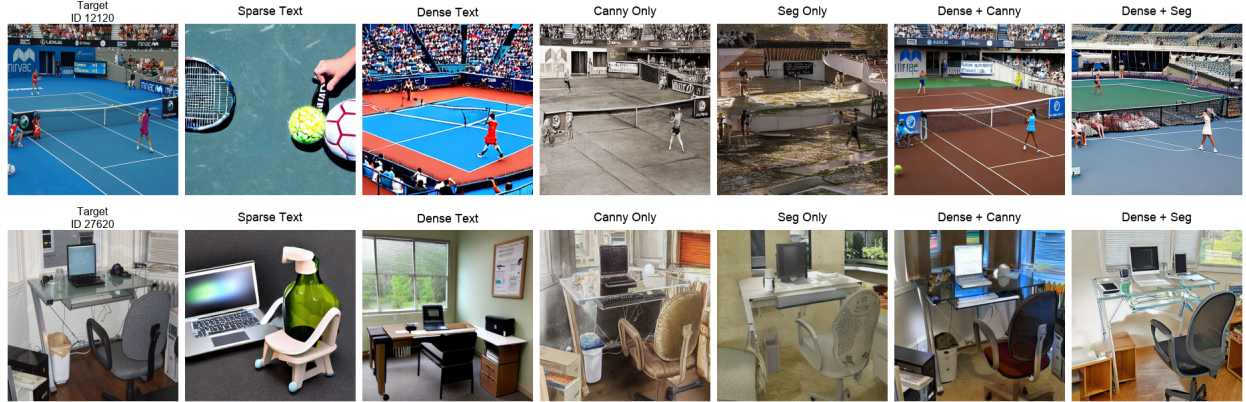


Figure 3: Qualitative comparison across conditioning modalities for the tennis-court scene (top row) and office scene (bottom row).

For the tennis-court scene (ID 12120), sparse text successfully captures the general concept of a tennis match but fails to reconstruct the overall scene layout, producing a close-up view of sports equipment rather than the full court. Dense text improves semantic alignment by generating a complete tennis court with players and spectators, but the exact spatial arrangement still differs from the target image. Structural conditioning preserves scene geometry more effectively. Both Canny-only and segmentation-only conditioning recover the court boundaries and overall layout, although they often distort object appearance and texture. Among all methods, Dense+Canny produces the closest reconstruction to the target, preserving both the semantic content of the scene and the geometric structure of the court. Dense+Segmentation also reconstructs the scene successfully but exhibits less accurate court geometry and object placement.

For the office scene (ID 27620), text-only conditioning again struggles to preserve the target layout. Sparse text generates unrelated objects and fails to reconstruct the office environment, while dense text captures the general office concept but changes the arrangement of furniture and room structure. In contrast, both structural modalities correctly recover the desk, chair, computer monitor, and overall room geometry. Dense+Canny and Dense+Segmentation further improve realism by combining structural constraints with semantic information from the caption. Dense+Canny produces the most faithful reconstruction overall, while Dense+Segmentation preserves the major objects but introduces larger appearance differences.

Several trends emerge consistently across both examples and our broader visual inspections, strongly confirming that our quantitative metrics mirror human qualitative judgment. First, text-only conditioning primarily preserves semantic information but provides weak spatial guidance. Second, structural conditioning significantly improves reconstruction of scene layout and object placement. Third, Canny conditioning generally preserves fine geometric details more effectively than segmentation conditioning, which provides only coarse region-level information. Finally, combining semantic and structural conditioning yields the strongest reconstructions, supporting the quantitative results presented in Tables 2–4.

Despite the improvements provided by multimodal conditioning, appearance discrepancies remain visible. Colors, textures, and fine object details often differ from the target image even when the overall scene structure is preserved. This behavior is expected because edge maps and segmentation masks encode spatial structure rather than appearance information, while textual descriptions

provide only limited visual detail. Consequently, the model preserves geometry more reliably than exact visual appearance.

3.8 Win-Rate Analysis

Average metrics summarize overall performance but do not indicate how consistently a method outperforms competing approaches on individual images. Furthermore, mean aggregates can be heavily skewed by a few extreme outliers. To address these limitations, we compute per-image win rates across LPIPS, CLIP-Score, and DreamSim, mitigating the impact of anomalous cases.

Our analysis shows that dense text outperforms sparse text in 86–89% of comparisons across CFG values, confirming that richer descriptions provide stronger semantic guidance. This consistency increases with spatial conditioning. Among single modalities, Canny conditioning wins roughly 60% of comparisons against dense text and segmentation alone. Similarly, for dual-modality settings, Dense + Canny leads by winning 73–77% of comparisons.

While these results show a clear dominance of Canny over segmentation (Canny $\gg$ Segmentation and Dense + Canny $\gg$ Dense+Segmentation), the triple-modal setting introduces crucial nuance. At CFG 7.5 and CN 1.0, the Dense + Canny + Segmentation configuration achieves the highest win rate at 49.5%, followed by Dense + Canny at 37.2%, and Dense + Segmentation at 13.3%.

This reveals that segmentation is still valuable when Canny edges are present. Though weak on its own, segmentation acts as an effective secondary branch that enforces macro region layout at moderate ControlNet scales. Ultimately, the three modalities are synergistic rather than redundant: text establishes semantics, Canny anchors fine boundaries, and segmentation secures the overall layout.

3.9 Interpretation

The reconstruction task should be interpreted as a controlled benchmark for measuring the information content of different conditioning modalities. Because the Canny and segmentation signals are extracted directly from the target image, the experiment does not evaluate how easily a human can create these inputs. Instead, it evaluates how effectively the diffusion model can utilize each modality once the conditioning signal is available.

Under this interpretation, the reconstruction metrics serve as proxies for the amount of semantic and structural information preserved by each modality. Better reconstruction indicates that the modality contains information that is useful for guiding generation toward the target scene.

This framing is important because our goal is not image generation or user interaction, but rather understanding which conditioning signals contain the strongest reconstructive guidance for diffusion models. The results therefore provide insight into the relative value of text, edges, and semantic layouts as conditioning modalities.

3.10 Best Configuration Recommendation

Based on the results above, Dense+Canny is the strongest evaluated two-modality configuration, while Dense+Segmentation also improves substantially over segmentation-only conditioning. These findings suggest that the three modalities provide complementary information: dense text contributes scene semantics, Canny edges preserve fine geometric boundaries, and segmentation maps provide object-level region layout.

Motivated by this complementarity, we evaluate a final combined configuration using Dense Text + Canny + Segmentation. For this final model, we tried CFG = 5.0 / 7.5 for text guidance and ControlNet conditioning scale = 1.0 / 1.5 for both Canny and segmentation. Our tests show that pushing the ControlNet scale to 1.5 in the triple-modal setup causes a structural-strength trade-off where CLIP-Score drops to 32.46, so a moderate scale of 1.0 is preferred.

This configuration is intended as a practical recommendation for reconstruction-oriented use of Stable Diffusion 1.5 with ControlNet. Rather than selecting a single modality, the final model combines semantic, geometric, and layout guidance to provide a richer conditioning signal.

Table 7: Final recommended configuration.

Component	Setting
Text Conditioning	Dense Caption
Spatial Conditioning 1	Canny Edge Map
Spatial Conditioning 2	Semantic Segmentation Mask
CFG Scale	7.5
Canny ControlNet Scale	1.0
Segmentation ControlNet Scale	1.0
LPIPS ↓	0.501
CLIP-Score ↑	34.33
DreamSim ↓	0.321

3.11 Extension: Style Editing Beyond Reconstruction

The reconstruction experiments identify Dense Text + Canny as the strongest 2-modality configuration for preserving the original scene.

However, this raises another question:

If Canny edges preserves the most information, are they also the best conditioning modality for style editing tasks?

To investigate this question, we conducted a small qualitative study using several representative editing tasks, including cartoon stylization, cozy-environment transformation, modern interior redesign, and photorealistic enhancement. For each task, we applied the same textual editing instruction while comparing Canny-edge conditioning against segmentation-map conditioning.

The results suggest that the answer depends strongly on the downstream objective. Canny conditioning preserves fine geometric structure and object boundaries, making it highly effective for reconstruction tasks where fidelity to the original image is desired. However, the same structural constraints can become overly restrictive during style transfer. Strong edge preservation limits the model’s ability to substantially alter shapes, layouts, textures, and visual appearance.

In contrast, segmentation conditioning preserves higher-level scene layout while allowing greater flexibility in object appearance and texture generation. Across multiple editing tasks, segmentation maps generally enabled larger stylistic changes while maintaining coherent scene structure. This behavior was particularly evident in cartoon stylization and interior redesign tasks, where segmentation-based conditioning produced stronger visual transformations without completely losing scene identity.

These observations indicate that the modality with the highest information capacity is not necessarily the most appropriate modality for every application. For reconstruction-oriented tasks, preserving as much structural information as possible is beneficial, making Canny-based conditioning highly effective. For creative editing tasks, however, a less restrictive conditioning signal may be preferable because it provides the diffusion model with greater freedom to realize the desired style transformation.

Consequently, our findings suggest a practical guideline for ControlNet users:

- Use Canny-edge conditioning when accurate reconstruction and geometric fidelity are the primary goals.
- Use segmentation conditioning when substantial stylistic modification is desired while preserving overall scene layout.
- Combine modalities when both fidelity and controllable editing are important.

This extension broadens the scope of the project beyond reconstruction benchmarking and demonstrates that modality selection should ultimately depend on the intended downstream task.

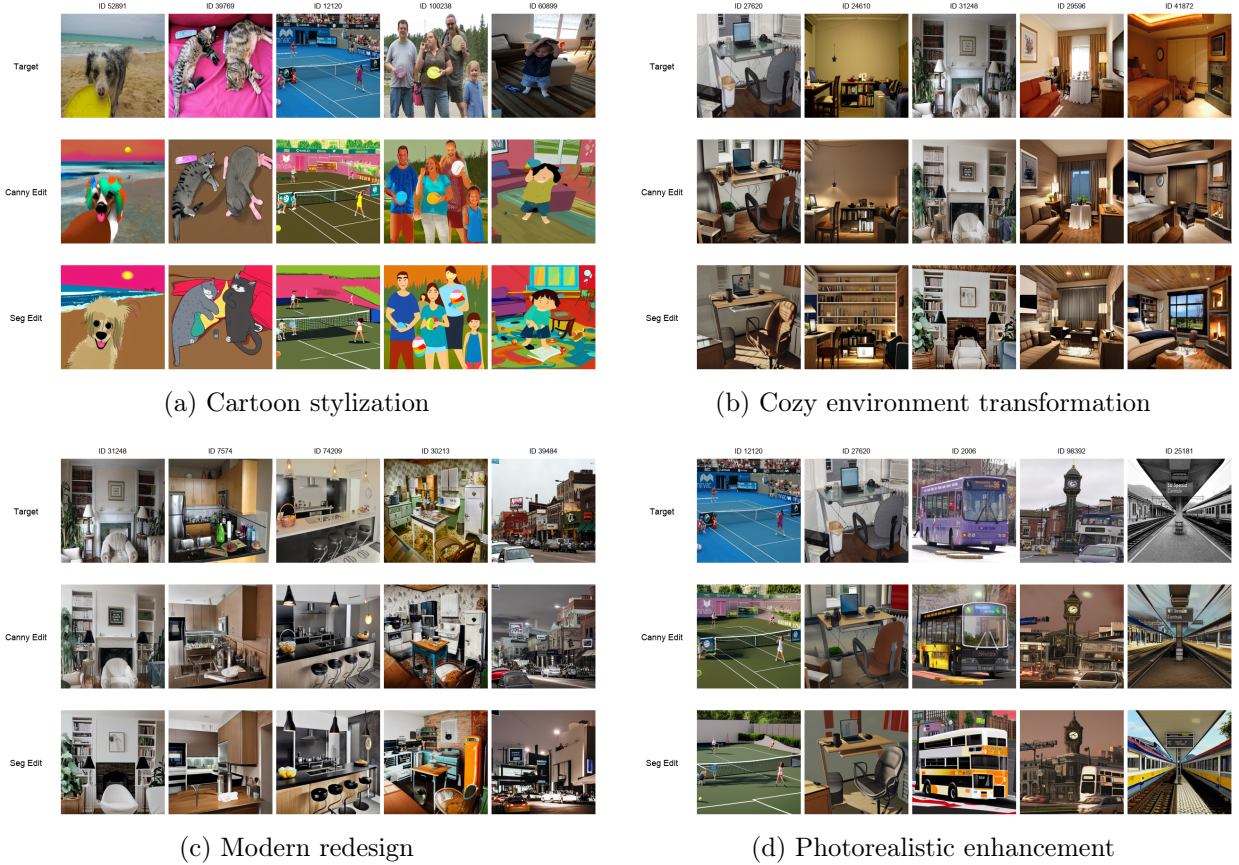


Figure 4: Style-editing comparison between Canny-edge conditioning and segmentation conditioning across four editing tasks.

3.12 Limitations and Future Work

While reconstruction provides a controlled benchmark for evaluating conditioning modalities, it does not directly measure human intent. In our experiments, Canny edge maps and segmentation masks are extracted from the target image itself, whereas real-world users typically provide imperfect sketches, layouts, or textual descriptions.

Additionally, all experiments are conducted using Stable Diffusion 1.5 and ControlNet. More recent diffusion architectures may utilize conditioning signals differently due to larger text encoders and improved multimodal integration.

Future work could investigate user-provided sketches, human-authored scene layouts, or multi-modal prompting workflows that more closely resemble practical content creation settings. Another promising direction is evaluating newer diffusion models such as SDXL or Flux under the same reconstruction benchmark.

4 Team Responsibilities

- **Sophia Huang:** dataset selection, dense caption construction, ControlNet Canny and segmentation pipeline integration, experiment execution, evaluation analysis.
- **Yixiao Zhang:** text-only implementation, evaluation pipeline, metric validation, data verification, experiment execution, comparative analysis, result aggregation.

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